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SIMPLE INTEREST

Department of Training & Placement

Aditya University

Introduction

- **Principal:**

  The money borrowed or lent out for a certain period is called the principal or the sum.

- **Interest:**

  Extra money paid for using other's money is called interest.

- **Simple Interest (S.I.):**

  If the interest on a sum borrowed for certain period is reckoned uniformly, then it is called simple interest.

Key Formulae

Let S.I = Simple Interest, Principal = P, Rate = R% per annum (p.a.) and Time = T years. Then

$$S.I = \frac{P \times T \times R}{100}$$

In order the calculate the total amount, the following formula is used:

Amount (A) = Principal (P) + Interest (I)

Where, **Amount** (A) is the total money paid back at the end of the time period for which it was borrowed.

Questions

Ques.1 Find the Simple Interest on ₹8,000 at 10% per annum for 3 years.

a) 2400

b) 2600

c) 1200

d) 1540

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Questions

Ques.2 A sum of money will double itself in 4 years at simple interest with a yearly rate of _____ %.

a) 12%

b) 25%

c) 20%

d) 35%

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Questions

Ques.3 A sum of Rs. 14,000 amounts to Rs. 22,400 in 12 years at the rate of simple interest. What is the rate of interest (in %)?

a) 8%

b) 10%

c) 5%

d) 6%

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Questions

Ques.4 What sum of money will produce Rs. 70 as simple interest in 4 years at 3 1/2 percent?

a)300

b)400

c)500

d)600

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Questions

Ques.5 The simple interest on a sum of money is one-fifth of the principal, and the number of years is five times the rate of interest. Find the rate of interest (in %).

a) 1.6%

b) 2%

c) 2.5%

d) 1.8%

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Questions

Ques.6 Ramesh borrowed Rs. 1200 at 15% rate of interest. He repaid Rs. 560 at the end of the first year. What is the amount paid by him at the end of the second year to discharge his loan which was calculated at simple interest?

a)1150

b)1200

c)1300

d)1250

Questions

Ques.7 A man took a loan from a bank at the rate of 16% p.a. simple interest. After 4 years, he had to pay Rs. 6400 as interest for the period. What was the principal amount borrowed by him?

a)8000

b)10000

c)12000

d)6400

Questions

Ques.8 How much time will it take for an amount of Rs.250 to yield Rs.100 as interest at the rate of 2.5% p.a. simple interest?

a)8.5

b)24

c)16

d)20

Questions

Ques.9 How much time will it take for an amount of Rs. 900 to yield Rs. 81 as interest at 4.5% per annum of simple interest?

a)1 year

b)2 years

c)3 years

d) 4 years

Questions

Ques.10 What sum will earn ₹1,500 as SI in 5 years at 6%?

a)4000

b)4500

c)5000

d)6000

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Questions

Ques.11 A sum of money at simple interest amounts to Rs. 815 in 3 years and to Rs. 854 in 4 years. The sum is _____.

a)690

b)700

c)698

d)654

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Questions

Ques.12 Raj bought $ 15,000 from a bank to buy a car at 10% simple Interest. If he paid $ 9,000 as interest while clearing the loan, find the time for which the loan was given.

a)4 years
b)5 years
c)6 years
d)7 years

Questions

QUES.13 Rakesh invested $ 10,000 in a savings bank account that earned 2% simple interest. Find the interest earned if the amount was kept in the bank for 4 years.

a)700
b)800
c)750
d)850

Questions

QUES.14 A sum fetched a total simple interest of Rs. 929.20 at the rate of 8% per annum in 5 years. What is the sum?

a)2323

b)1223

c)2000

d)2500

Questions

QUES.15 A man took loan from a bank at the rate of 8% p.a. simple interest. After 4 years he had to pay Rs. 6200 interest only for the period. The principal amount borrowed by him was _____.

a)16000

b)18000

c)20000

d)19375

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COMPOUND INTEREST

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Questions

Principal = P, Rate = R % per annum, Time = n years

1. Amount = $P\left[1+\frac{R}{100}\right]^{n}$ - - - - - - - - - [Interest compounded annually]

1. Amount = $P\left[1+\frac{(R/2)}{100}\right]^{2n}$ - - - - - - - - - [Interest compounded Half-yearly]

1. Amount = $P\left[1+\frac{(R/4)}{100}\right]^{4n}$ - - - - - - - - - [Interest compounded quarterly]

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Questions

Numerical on Population :

Latest Population of a Particular city → P2 = P1[1 + {R/100}]^n or P2 = P1[1 – {R/100}]^n

If rates are different for every year (R1,R2,R3 for 1,2,3 years respectively. Then,

Amount for n years = P [1 + {R1/100}] [1+ {R2/100}] [1+{R3/100}]…..

Questions

Present worth → P = X /[1+ {R/100}]^n]

X→ previous rate, R→ Rate of interest p.a, n →number of years

Difference between CI and SI

For 2 years

$$CI - SI = P(R/100)^2$$

For 3 years

$$CI - SI = P (R/100)^2 X [(300 + R)/100]$$

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Questions

Ques.1 Find the compound interest (CI) on Rs. 12,600 for 2 years at 10% per annum compounded annually.

a) 2646

b) 2556

c) 3123

d) None of these

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Questions

Ques.2 Find the compound interest on Rs 48,000 for one year at 8% per annum when compounded half-yearly.

a) 2400

b) 2600

c) 1200

d) None of these

Questions

Ques.3 Find the compound interest on Rs. 8000 at 15% per annum for 2 years 4 months, compounded annually.

a) 2436

b) 2639

c) 1220

d) 3109

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Questions

Ques.4 The compound interest on Rs. 30,000 at 7% per annum is Rs. 4347. The period (in years) is:

a) 2

b) 2.5

c) 3

d) 4

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Questions

Ques.5 A bank offers 5% compound interest calculated on half-yearly basis. A customer deposits Rs. 1600 each on 1st January and 1st July of a year. At the end of the year, the amount he would have gained by way of interest is:

a) Rs.120

b) Rs.121

c) Rs.122

d) Rs.123

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Questions

Ques.6 What is the difference between the compound interests on Rs.5000 for one and half years at 4% per annum compounded yearly and half-yearly?

a) 2.04

b) 3.06

c) 4.08

d) 8

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Questions

Ques.7 The difference between S.I and C.I compounded annually on a certain sum of money for 3 years at 10%p.a is Rs. 15.50. find the PRINCIPAL?

a) 650

b) 500

c) 525

d) 625

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Questions

Ques The difference between S.I and C.I compounded annually on a certain sum of money for 2 years at 4%p.a is Rs. 1. find the PRINCIPAL?

a) 600

b) 645

c) 525

d) 625

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Thank You